

SilverHands — Complete Project Structure

India's AI-Powered Digital Livelihood Platform for Senior Citizens & Homemakers

Track 3 — Hexaware Premier League (HPL) Mavericks Hackathon Stack: React · Django · RabbitMQ · MongoDB · Docker · n8n

1. One-Line Pitch

SilverHands turns lifelong skills into livelihood — through a mesh of AI agents that onboard, verify, match, promote, and grow every seller automatically, wrapped in a voice-first interface built for people who've never used a marketplace app before.

2. Core Differentiators (say these explicitly to judges)

- 1. **Agent Mesh, not a chatbot** — 8 specialized AI agents communicating over RabbitMQ, not one LLM call bolted onto a form.
 - 2. **Trust-first onboarding** — Aadhaar verification happens *before* anything else, so every listing on the platform is backed by a real, verified identity from day one.
 - 3. **Gamified livelihood, not just a listing** — leaderboards, badges, and progressive verification tiers turn "posting a product" into a rewarding habit loop — designed deliberately for a demographic that responds well to recognition and routine.
 - 4. **Every seller is a content creator** — the platform auto-generates their social posts, ads, and promotional content, so a 65-year-old pickle-maker gets an Instagram presence without knowing what Instagram is.
 - 5. **n8n as the automation backbone** — live operational visibility (new sellers, verifications, orders, flagged accounts) pushed straight into a Google Sheet in Drive, so hackathon judges and admins can watch the platform work in real time — no dashboard build needed.
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3. User Roles

Role	Description
Seller	Senior/homemaker offering skills, services, or products
Buyer	Local customer seeking services/products
Guide	A family member linked to a seller's account — can help set up, monitor orders, and receive alerts on the seller's behalf (opt-in, seller stays in control)
Admin	Platform ops — sees n8n-fed live dashboard

4. The Agent Mesh — 8 Agents over RabbitMQ

Each agent is a Django service/worker with its own queue. Events flow through RabbitMQ so agents work asynchronously and independently — this is your strongest technical story for judges.

#	Agent	Trigger Queue	What it does
1	Verification Agent	<code>user.registered</code>	Runs Aadhaar OCR + validation (via DigiLocker/UIDAI sandbox API or mocked verification for demo), face-match against Aadhaar photo, marks account <code>verified_basic</code>
2	Onboarding Agent	<code>user.verified</code>	Converts voice/text/photo skill description → structured profile (skill tags, category, auto-written bio, suggested pricing)
3	Matchmaking Agent	<code>profile.updated</code> , <code>request.created</code>	Embedding-based similarity match on skill + location + language; proactively pushes leads to sellers instead of requiring search
4	Content/Promotion Agent	<code>product.listed</code>	Auto-generates social captions, poster images (template + AI), and ad copy; publishes to <code>content.ready</code> for the Marketing layer to push out
5	Trend & Demand Agent	<code>scheduled</code> (weekly)	Analyzes seasonal demand + social trend signals, pushes "make more of X this month" recommendations to sellers
6	Delivery Orchestration Agent	<code>order.confirmed</code>	Auto-requests a rider via Rapido API (or partner logistics API) when order is ready for pickup, tracks status back to buyer/seller
7	Gamification Agent	<code>order.completed</code> , <code>review.posted</code>	Updates points, badges, leaderboard rank, checks if seller has hit the order-count threshold to unlock next verification tier
8	Voice/Accessibility Agent	<code>persistent WebSocket</code>	Handles all voice commands app-wide ("read my orders," "post this," "how much did I earn") — not just voice search

n8n sits alongside this mesh, subscribed to key events (`user.verified`, `order.completed`, `flag.raised`, `product.listed`) and writes structured rows live into a Google Sheet in Drive — your judge-facing "operations room" view, built with zero custom dashboard code.

5. Aadhaar Verification Flow (Step 1, before anything else)

1. User signs up → prompted immediately for Aadhaar (via DigiLocker consent flow, or a mocked verification service for hackathon scope)
2. OCR + validation confirms name/DOB match
3. Optional liveness/face-match selfie for extra trust
4. Account marked `verified_basic` — **only verified users can list products/services**
5. Progressive trust tiers unlock as sellers complete orders (see gamification below) — so verification isn't just a one-time gate, it's a growth path

This directly satisfies the "Verified user" and "verification shld be done in the start" requirements, and doubles as your responsible-AI/safety talking point.

6. Gamification System — "Oldies are just old babies" 🎯

Mechanic	Detail
Points	Earned per order completed, per positive review, per profile completeness
Leaderboard	Weekly + all-time, filterable by category (Tailoring, Cooking, Tutoring, etc.) and by locality — local pride matters more than national ranking for this audience
Verification Tiers	<code>New Seller</code> → complete 5 orders → <code>Trusted Seller</code> (badge + higher search ranking) → complete 20 orders → <code>Master Craftsperson</code> (featured placement + eligible for collaborative business invites)
Badges	"First Sale," "5-Star Streak," "Festival Special" (seasonal), "Community Mentor" (helped onboard another senior)
Streaks	Weekly activity streak — nudged gently via voice notification, not aggressive push spam

7. Collaborative Business (Pairing Sellers)

- Sellers can form a "Co-op Listing" — e.g., one does tailoring, another does embroidery finishing → jointly list a combined product/service
- Matchmaking Agent suggests pairings based on complementary skill tags + proximity
- Revenue split handled automatically per an agreed ratio set at pairing time

- Great pitch line: *"We don't just create sellers, we create local business partnerships."*
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8. Family "Guide" Role

- A senior can invite a family member as a **Guide** via a simple link/OTP
 - Guide gets a read-only + assist dashboard: can help write listings, view orders, get alerts — but cannot act as the seller or take payments
 - Positions the platform as intergenerational, not isolating — strong social-impact narrative
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9. Content Creator Layer

- Every listing auto-generates: an Instagram/Facebook-ready caption + hashtags, a simple branded poster (template + product photo), and a short auto-narrated video slideshow (optional stretch goal)
 - Seller taps **one button** — "Promote this" — content agent handles the rest and can auto-post via connected social accounts, or export for the seller/guide to post manually
 - This is your strongest "wow" feature for a demo: type/speak a product description → 10 seconds later, a ready-to-post Instagram ad exists
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10. Delivery Integration (Rapido-style)

- When an order is confirmed, the **Delivery Orchestration Agent** calls a rider-partner API automatically
 - Seller doesn't need to arrange pickup — a rider is requested the moment payment/order is confirmed
 - Live tracking surfaced to both buyer and seller in familiar map UI
 - For hackathon scope: integrate a rider API if available, otherwise mock the flow convincingly with realistic status transitions (Requested → Rider Assigned → Picked Up → Delivered)
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11. Local & Nearest Discovery

- Location-aware feed: "Sellers near you," "Buyers near you," "Trending in your area"
 - Nearest-first sorting on all discovery screens by default (senior users shouldn't have to filter manually)
 - Map view for buyers browsing local sellers/products
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12. Senior-Friendly UX (non-negotiable specifics)

- Voice-first: persistent mic button, same position on every screen

- No mandatory typing anywhere — voice + photo covers 100% of onboarding
- Large tap targets (48px+), max 2 primary actions per screen
- WhatsApp-style chat/list UI patterns (familiar mental model)
- Full regional language support — auto-detected from first voice input, not a settings toggle
- Read-aloud confirmation before any payment or public listing goes live
- Offline-first PWA — actions queue locally and sync when back online

13. n8n Automation Backbone

Example flows to build/demo:

n8n Flow	Trigger	Action
Live Ops Sheet	Any key RabbitMQ event (webhook)	Append row to Google Sheet: event type, user, timestamp, status
New Seller Alert	<code>user.verified</code>	Row in "Sellers" tab + optional Slack/email ping to admin
Flagged Account	<code>flag.raised</code> (from Trust agent)	Row in "Review Queue" tab, highlighted red, admin notified
Weekly Trend Digest	Scheduled	Pulls demand data → writes summary to "Trends" tab → triggers Trend Agent to notify sellers
Order Funnel Tracking	<code>order.created</code> → <code>order.completed</code>	Updates a live "Orders" tab so judges can watch conversion happen during the demo

This is a big value-add for a hackathon demo: **judges can literally open a live Google Sheet and watch platform activity update in real time** while you demo the app — no custom admin panel needed, and it screams "automation-first."

14. MVP Scope (build this for the hackathon — cut everything else)

Must-build (Day 1 priority):

1. Voice/text + photo onboarding → AI-generated profile (Onboarding Agent)
2. Aadhaar verification flow (can be mocked/sandboxed for demo)
3. Product/service listing with AI-generated bio + suggested price
4. Buyer-side discovery feed with location sort
5. Basic matchmaking (skill + location) — even a simple scored query counts as "AI matching" if framed well

6. Order flow: request → confirm → complete (mock payment is fine)
7. Gamification: points + one leaderboard + 2-3 badges
8. n8n → Google Sheet live ops flow (this is cheap to build and huge for demo impact)
9. Voice navigation for at least 3 core actions (post listing, check orders, read notifications)

Stretch goals (build if time remains, in this order):

1. Content/Promotion Agent (auto social post generation) — highest "wow" per hour of effort
2. Delivery Agent (Rapido-style mock flow)
3. Family Guide role
4. Collaborative business pairing
5. Trend & Demand Agent (weekly recommendations)
6. Real social auto-posting integration

Explicitly cut for hackathon timeframe: real payment gateway, real Rapido production API (mock it), multi-language voice for every screen (pick 2 languages max for demo), production-grade Aadhaar/UIDAI integration (use DigiLocker sandbox or a convincing mock).

15. Suggested Repo/Service Structure

```
silverhands/
├── frontend/                                # React
│   ├── src/
│   │   ├── components/
│   │   ├── screens/ (Onboarding, Discover, MyListings, Orders, Leaderboard)
│   │   └── voice/          # voice command handler, TTS/STT hooks
├── backend/                            # Django
│   ├── core/                        # users, auth, guide-linking
│   ├── verification/              # Aadhaar OCR/validation service
│   ├── listings/                  # products/services, pricing
│   ├── matchmaking/               # embedding search, recommendation logic
│   ├── orders/                    # order lifecycle, delivery hooks
│   ├── gamification/              # points, badges, leaderboard
│   └── agents/                    # RabbitMQ consumers per agent
├── automation/
│   └── n8n-flows/                  # exported n8n workflow JSONs
├── infra/
│   ├── docker-compose.yml         # Django, React, MongoDB, RabbitMQ, n8n
│   └── rabbitmq/                  # queue/exchange definitions
└── docs/
    └── architecture-diagram.png
```

16. Demo Script (5-7 min, this is what wins hackathons)

1. **(30s)** Open with the problem — a real stat on India's ageing population + idle homemaker skill pool
2. **(1 min)** Live: a "senior" speaks in Tamil/Hindi about their skill → watch AI generate full profile + listing in real time
3. **(30s)** Show Aadhaar verification step — emphasize trust-first design
4. **(1 min)** Switch to buyer view → show proactive AI match notification ("3 buyers near you want X")
5. **(1 min)** Tap "Promote this" → show auto-generated Instagram-ready post appear instantly
6. **(1 min)** Complete an order → show delivery agent auto-requesting a rider, leaderboard points update live
7. **(1 min)** **Pull up the live Google Sheet fed by n8n** — every action just performed is already logged there in real time
8. **(30s)** Close on impact: financial independence, digital inclusion, preserved traditional skills, intergenerational connection

Want me to start scaffolding actual code next — the Django agent structure with RabbitMQ consumers, the docker-compose for the full stack, or the n8n workflow JSON for the live Google Sheet automation?